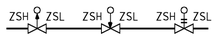
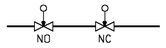
ON / OFF VALVES

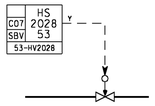


1) Fail open, 2) fail close and 3) stay put upon
loss of electrical signal from SAS, respectively.

ZSH: "OPEN" limit switch, feedback signal to SAS, if applicable
ZSL: "CLOSED" limit switch, feedback signal to SAS, if applicable

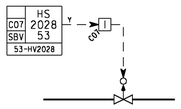


Normally Open / Normally Closed: The conception "normal" tells whether
the valve is open or closed most of the time when the process is in operation.
(It does not tell anything about the physical properties of the valve itself,
and it does not affect SAS configuration.)



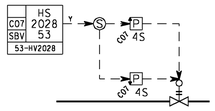
FC (Fail Close) on-off valves will close if the electrical signal from SAS is lost.
A low control signal from SAS (0 V DC) will close the valve.

Terminal of FB:	Hardwired signal tags:	
Y	53-HY2028	Open / close signal from SAS



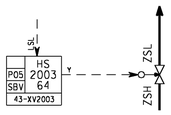
FO (Fail Open) on-off valves will open if the electrical signal from SAS is lost.
A low control signal from SAS (0 V DC) will open the valve.

Terminal of FB:	Hardwired signal tags:	
Y	53-HY2028	Open / close signal from SAS



FM (Fail Maintain) on-off valves have one solenoid for opening
and one solenoid for closing of the valve.

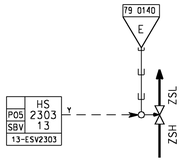
Terminal of FB:	Hardwired signal tags:	
Y	53-HY2028	Open / close signal from SAS



Process shutdown valve (XV).

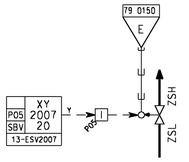
The valve can be operated manually from OS, or shut down by the PSD node.
Shutdown by means of LSL is latched and has to be reset by operator.

Terminal of FB:	Hardwired signal tags:	
Y	43-XY2003	Open / close signal from PSD node
XGL	43-ZSL2003	Position Low (closed) feedback to PSD
XGH	43-ZSH2003	Position High (open) feedback to PSD

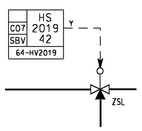


Emergency shutdown valve (ESV).
The valve has two solenoids: one connected to the ESD node, and the other to the
PSD node. Manual operation from OS is done by means of the PSD node.
The valve is closed by the ESD node and solenoid when stated ESD level is released.
After ESD shutdown, the valve is latched, and has to be reset locally.
The valve also has a PSD purpose. If it is closed by PSD, it is reset the same way
as an XV-valve. The valve has two limit switches connected to the PSD node.
When the valve is closed by ESD, the ESD node shall send an internode signal
to the PSD node to close the valve from PSD additionally,
to avoid travel fault in PSD.

Terminal of FB:	Hardwired signal tags:	
Y	13-XY2303	Open / close signal from PSD node
XGL	13-ZSL2303	Position Low (closed) feedback to PSD
XGH	13-ZSH2303	Position High (open) feedback to PSD

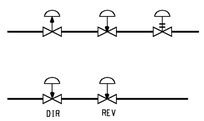


Blow down valve (ESV).
The functionality is the same as for other ESV valves, except that
the valve shall open when the stated ESD-level (blowdown) is released, and
the valve is FO.



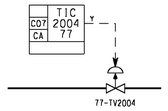
3 way solenoid valve. Fail position: medium can then flow between the two
process lines which are connected to the white parts of the valve.
A high control signal from SAS (Y = 1) will make the valve open to the
process line which is connected to the black part of the valve, and
close one of the other. The ZSL-feedback is active when the valve
is closed to the process line which is connected to the black part of the valve.

ANALOG VALVES



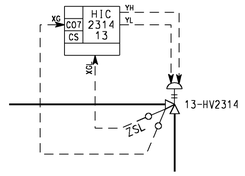
Analog / modulating valves: 1) Fail open, 2) fail close and 3) stay put
upon loss of signal (deenergized) from SAS, respectively.

Direct: The valve opens gradually with increasing electrical output control signal from SAS.
Reverse: The valve closes gradually with increasing electrical output control signal from SAS.
DIR/REV does not depend on whether the valve is FC/FO.
If not indicated, DIR can be assumed to apply.



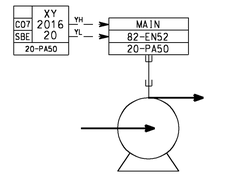
Example

CHOKE VALVE



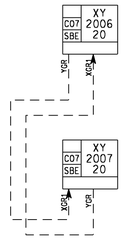
The HIC provides two outputs: one a pulse to increase the opening of the valve a bit,
and the other to reduce the opening a bit. The production valve at Huldra needs 192
pulses to open from 0 - 100 %. The operator should be able to control the valve position
by increments of 0.5 %.
The 24VDC ON-pulse duration is to be minimum 1.25 s, and the 24VDC OFF-pulse duration
minimum 1.25 s, i.e. the total pulse period is to be minimum 2.5 s (irrespective of
whether the valve-opening is increased or reduced). Thus the total stroke time is
ca eight minutes. However, it shall be possible for the operator to adjust the speed
of the valve opening, allowing opening stroke time of up to eight hours.

Terminal of FB:	Hardwired signal tag:	
XGL	13-ZSL2314	Valve closed feedback
XG	13-ZT2314	Valve position feedback
YH	13-HY2314A	Pulse open valve, command from SAS
YL	13-HY2314B	Pulse close valve, command from SAS



Electrical on/off units controlled by SAS
(Motor, heater, fan etc.)

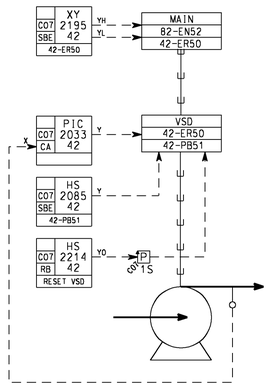
Terminal of FB:	Bus signal tags:	
XGH	20-XY2016R	Running feedback
XE	20-XY2016A	Available feedback
YH	20-XY2016H	Pulsed Start signal from PCS
YL	20-XY2016L	Pulsed Stop signal from PCS



Automatic DUTY-STANDBY functionality between two units, e.g. motors.

Both FBs, controlling one unit each, must be set in AUTO beforehand, and
one unit must be chosen to have priority 1 (duty unit) and
the other to have priority 2 (standby unit).

The standby unit will start automatically
if the duty unit fails.



Variable speed drive (VSD)
An example from Huldra project:

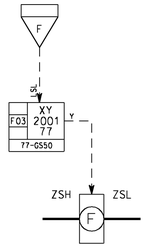
Terminal of FB (SBE):	Bus signal tags:	
XGH	42-XY2195R	Closed feedback from VSD-feeder (690V contactor)
XE	42-XY2195A	Available signal from MCC
YH	42-XY2195H	Pulsed Close signal to VSD-feeder
YL	42-XY2195L	Pulsed Open signal to VSD-feeder

Terminal of FB (CA):	Hardwired signal:	
X	42-PT2033	Analog feedback of process value to SAS
Y	42-PY2033	Analog output control value to VSD

Terminal of FB (SBE):	Hardwired signal:	
XGH	42-XY2085R	Running feedback from VSD/pump
Y	42-XY2085H	Steady start/stop signal to VSD/pump
XE	NA	Available for VSD/pump is set constantly to TRUE (logical "1") in logic

Terminal of FB (RB):	Hardwired signal:	
YO	42-HS2214	Reset VSD from OS

HVAC



Fire dampers are operated by the F&G node,
using the functionblock SB_FG.

Hardwired signal tags for the field interface: SAS - fire damper:

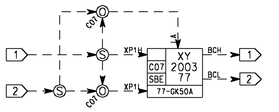
Y (Start / Stop from F&G):	77-XY2001
ZSL ("Closed" feedback from fire damper):	77-ZSL2001
ZSH ("Open" feedback from fire damper):	77-ZSH2001

Unconventional naming of the terminals for the SB_FG:

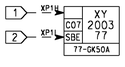
Conventional naming:	Real naming for the SB_FG:
LSL	RX
XH	XH
XL	RX
YF	YF
BCL	BCH
BCH	BCL
Y	YL

Using the conventional terminal naming on the SCD
the readability should be maintained.

The logic in the F&G node is generally not reflected at the SCD's, but
concerning HVAC the F&G-logic is shown (to some extent), due to the
close co-operation with the PCS-logic.



To save space at the SCD, the structure
to the left is shown as the structure
to the right.



Rev.	Date	Description	Made by	Disc. check	Disc. appr.	Contr. appr.	Comp. appr.
A1	30.09.01	AS BUILT	EiL				GS
0	31.03.01	ISSUED FOR CONSTRUCTION	SNo				
03D	10.03.00	RE-ISSUED FOR DESIGN	DD				
02D	26.11.99	ISSUED FOR DESIGN	SNo				
01R	25.10.99	ISSUED FOR REVIEW	SNo				

Acceptance Code: 1☐ 1B☐ 2A☐ 2☐ 3☐ 4☐ Date: Sign:

STATOIL
Kværner Oil & Gas

Project:
HULDRA EPC WHPT

Drawing title:
SYSTEM CONTROL DIAGRAM
LEGEND 5, ACTUATORS

N/A	
Tag no.	
	NTS
SDL Code	Scale
P.O.No : 480018-P01-01	A1
Projection	Dwg. Format
Siemens Doc. No :	00
Area Code	Syst. Code
Drawing No : C025-W-H000-J-*E-014-01	1of1 A1
Project install / Origin. / Disc./Doc.Type / Seq.No.	Sheet / Rev.